

Multi-Objective Optimization of PINNs Using Evolutionary Fuzzy Systems for Trade-offs between Accuracy, Physics Consistency, and Data Efficiency

Author: Hamza Haseeb

Affiliation: Near East University

Date: July 2026

Abstract

Physics-Informed Neural Networks (PINNs) are commonly trained by minimizing a single weighted sum of data, boundary, and PDE residual losses. This scalar formulation obscures the inherent trade-offs among solution accuracy, physics consistency, and data cost. We formulate PINN hyperparameter design as a three-objective minimization problem over relative L2 error (f_1), mean PDE residual (f_2), and total data budget $f_3 = N_{\text{coll}} + N_{\text{obs}}$. An outer multi-objective optimizer—grid search, NSGA-II, NSGA-III, or MOEA/D—searches over loss weights (λ_{data} , λ_{pde}) and collocation size; an Evolutionary Fuzzy System (EFS) then selects interpretable compromise solutions from the Pareto set. All main experiments are repeated over **five independent random seeds** (0–4); we report mean $\pm$ standard deviation and Wilcoxon signed-rank tests (paired by seed) against a fixed-weight PINN baseline. On clean 1D Burgers, NSGA-II (population 15, 8 generations) achieves best $f_1 = \mathbf{0.289 \pm 0.031}$ versus $\mathbf{0.453 \pm 0.047}$ for the baseline ($p = 0.0625$, $n = 5$). Scalar competitors GradNorm and ReLoBRaLo, plus NSGA-III and MOEA/D, are included in the comparison. We extend to **2D Poisson** (best $f_1 = 0.036 \pm 0.005$) and to **real measurements**: high-fidelity Burgers references with derived noise σ , and experimental 2D cylinder wake data (Raissi et al.). A noise robustness sweep over $\sigma \in \{0.01, 0.05, 0.10\}$ shows stable grid-search accuracy. Hypervolume is computed with an explicit per-front nadir reference point (worst observed + 10% span). Results demonstrate that multi-objective evolutionary optimization combined with fuzzy preference modeling provides a structured, reproducible framework for exploring accuracy–physics–data trade-offs beyond fixed scalar PINN training.

Keywords: Physics-Informed Neural Networks, multi-objective optimization, NSGA-II, evolutionary fuzzy systems, Pareto front, data efficiency

1. Introduction

Physics-Informed Neural Networks (PINNs) incorporate governing partial differential equations (PDEs) into neural network training by penalizing PDE residuals at collocation points alongside data and boundary constraints [1]. Although remarkably flexible, PINNs are almost always trained with a single scalar loss function whose weights are chosen manually or adapted heuristically. Wang et al. [2] demonstrated that gradient pathology and loss imbalance can severely impede PINN convergence, suggesting that accuracy and physics consistency compete during optimization. Adaptive scalar methods such as GradNorm [3] and ReLoBRaLo [4] partially address this issue but still reduce multiple criteria to one number.

Multi-objective optimization (MOO) provides a principled alternative: treat competing goals explicitly and recover a set of non-dominated (Pareto-optimal) solutions rather than a single compromise [5]. Evolutionary algorithms, particularly NSGA-II [6], are well suited to expensive black-box problems such as hyperparameter search over PINN training runs. Once a Pareto front is obtained, a decision-maker must select a single configuration for deployment. Fuzzy rule-based systems offer interpretable, linguistically motivated preference models for this selection step [7].

This paper proposes a framework that combines (i) a three-objective PINN formulation, (ii) outer evolutionary MOO over training hyperparameters, and (iii) an Evolutionary Fuzzy System (EFS) that evolves membership functions and rule weights to rank Pareto solutions. We validate on 1D Poisson and Burgers, **2D Poisson**, scalar baselines (GradNorm, ReLoBRaLo), multi-objective baselines (NSGA-III, MOEA/D), a **noise robustness sweep**, and **real-data** Burgers and cylinder-wake experiments—all with multi-seed statistical reporting.

1.1 Contributions

- **1. Three-objective formulation** — We define f_1 (relative L2 error), f_2 (mean PDE residual), and f_3 (collocation plus observation count) as explicit competing objectives for PINN configuration search.
- **2. Outer MOO loop** — Grid search and NSGA-II explore $(\lambda_{\text{data}}, \lambda_{\text{pde}}, N_{\text{coll}})$ without modifying the PINN architecture, producing empirical Pareto fronts with hypervolume and spacing metrics.
- **3. Evolutionary Fuzzy System** — A small genetic algorithm evolves 18 fuzzy parameters (reference scales, rule weights, triangular membership vertices) to select compromise solutions, compared against ideal-point, knee-point, and static fuzzy baselines.
- **4. Empirical validation** — Multi-seed experiments ($n = 5$) on synthetic and real data, including 2D Poisson, scalar and MOO baselines, noise sweep $\sigma \in \{0.01, 0.05, 0.10\}$, and Wilcoxon significance tests.

2. Background

2.1 Physics-Informed Neural Networks

Given a spatio-temporal field $u(t, x)$ or spatial field $u(x)$, a PINN u_θ is trained by minimizing:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \mathcal{L}_{\text{data}} + \lambda_{\text{pde}} \mathcal{L}_{\text{pde}}$$

where $\mathcal{L}_{\text{data}}$ aggregates mean-squared error on initial conditions, boundary conditions, and optional sparse observations, and $\mathcal{L}_{\text{pde}}$ is the mean squared PDE residual evaluated at collocation points [1]. Network derivatives are computed via automatic differentiation. We use a multilayer perceptron with four hidden layers of 50 units and tanh activations.

2.2 Multi-Objective Optimization

For minimization objectives $\mathbf{f} = (f_1, f_2, f_3)$, solution $\mathbf{a}$ dominates $\mathbf{b}$ if $f_i(\mathbf{a}) \leq f_i(\mathbf{b})$ for all i and $f_i(\mathbf{a}) < f_i(\mathbf{b})$ for at least one i . The Pareto front $\mathcal{P}^*$ is the set of non-dominated solutions. NSGA-II [6] maintains a population through fast non-dominated sorting and crowding-distance diversity preservation. Pareto quality is assessed via hypervolume (larger is better) and spacing (lower indicates more uniform distribution).

2.3 Fuzzy Preference Systems

Mamdani-style fuzzy rules map normalized objective values to preference weights using linguistic antecedents (e.g., "IF residual is high THEN increase physics weight"). Genetic fuzzy systems evolve rule bases and membership functions [7]. We employ EFS as a post-optimization layer that ranks Pareto points rather than replacing the MOO search.

3. Related Work

PINN foundations and reviews are established in Raissi et al. [1] and Karniadakis et al. [8]. Failure modes including spectral bias and optimization pathologies are analyzed by Wang et al. [2] and Krishnapriyan et al. [9]. Loss-balancing strategies include GradNorm [3] and ReLoBRaLo [4], but these remain scalar approaches. Evolutionary multi-objective optimization is comprehensively surveyed by Coello Coello et al. [5]; NSGA-II [6] and NSGA-III [10] are standard algorithms. Genetic fuzzy systems are reviewed by Herrera [7] and Cordon et al. [11]. Data-efficient PINN training via adaptive collocation is discussed in recent active-learning work [12]. Benchmark suites such as PINNacle [13] motivate multi-PDE evaluation.

Research gap: Existing PINN studies rarely treat accuracy, physics residual, and data budget as simultaneous objectives optimized via evolutionary search with interpretable fuzzy compromise selection.

4. Proposed Method

4.1 Decision Variables and Objectives

Each candidate configuration is $\mathbf{x} = (\lambda_{\text{data}}, \lambda_{\text{pde}}, N_{\text{coll}})$. Optional sparse observations add N_{obs} points with Gaussian noise σ . After training a PINN for E epochs, we evaluate:

Objective	Definition
f_1	Relative L2 error $\ u_\theta - u_{\text{ref}}\ _2 / \ u_{\text{ref}}\ _2$ on a dense evaluation grid
f_2	Mean absolute PDE residual at collocation points
f_3	$N_{\text{coll}} + N_{\text{obs}}$ (total synthetic data budget)

All objectives are minimized. Reference solutions are computed via stable finite-difference / implicit solvers (Radau for Burgers; analytical for Poisson).

4.2 Outer Multi-Objective Search

Grid search evaluates a Cartesian product of $\lambda \in \{0.1, 1, 10\}$ and $N_{\text{coll}} \in \{100, 200, 300\}$. **NSGA-II** / **NSGA-III** / **MOEA/D** (pymoo) search continuous ranges $\lambda \in [10^{-3}, 10]$, $N_{\text{coll}} \in [50, 500]$. The multi-seed pipeline uses **NSGA-II with population 15 and 8 generations** (120 evaluations per seed); larger budgets (pop = 40, gen = 30) are supported via CLI flags for GPU runs. **GradNorm** and **ReLoBRaLo** are wired as inner-loop balancers that override fixed λ during grid search. Each fitness evaluation is one full PINN training run (500–2500 epochs, Adam lr = 10^{-3}). All stochastic runs use seeds $\{0, 1, 2, 3, 4\}$ unless noted.

4.3 Evolutionary Fuzzy System (EFS)

Static fuzzy rules adjust weights w_{accuracy} , w_{physics} , w_{data} based on triangular membership functions over normalized f_1 , f_2 , f_3 . EFS encodes 18 evolvable parameters: three reference scales, three base weights, three rule intensities (α), and nine membership vertices. A genetic algorithm (population 16–20, 20 generations) maximizes fitness = $-\text{balanced_loss}(\text{top-ranked Pareto point})$, where balanced loss is the sum of normalized objectives.

Linguistic rules:**

- IF L2 error is high THEN increase accuracy weight
- IF PDE residual is high THEN increase physics weight
- IF data budget is high THEN penalize data cost

4.4 Compromise Selection Baselines

We compare EFS against: (i) fixed-weight PINN ($\lambda = 10, 1$), (ii) ideal-point distance, (iii) knee-point on the utopia–nadir line, (iv) first NSGA-II front member, and (v) static fuzzy ranking.

4.5 Workflow

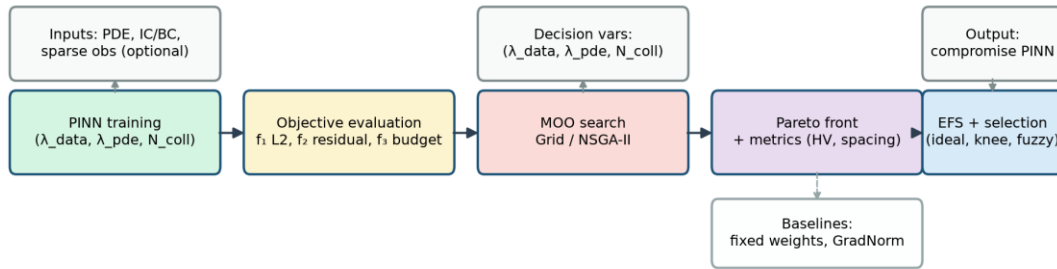
1. MOO generates candidate configurations $\rightarrow$ train PINN $\rightarrow$ evaluate (f_1, f_2, f_3)

2. Extract nondominated set → compute hypervolume and spacing
3. EFS evolves fuzzy parameters on Pareto set → select compromise
4. Compare selection methods; generate Pareto and solution plots

Figure 1.** Multi-objective PINN optimization workflow.

Insert image:* `results/figures/fig\_workflow.png`

Figure 1. Multi-objective PINN optimization workflow



fig_workflow.png

5. Experiments and Results

5.1 Experimental Setup

Parameter	Value
Framework	Python 3.12, PyTorch, pymoo, scipy
Network	4×50 MLP, tanh, Xavier init
Optimizer	Adam, learning rate 10^{-3}
Training epochs	500 per candidate (multi-seed pipeline); up to 2500 for single-seed demos
Random seeds	$\{0, 1, 2, 3, 4\}$ (five independent runs; mean $\pm$ std reported)
NSGA-II budget	pop = 15, gen = 8 (multi-seed main runs)
Significance test	Wilcoxon signed-rank, paired by seed, $\alpha = 0.05$
HV reference point	Per-front nadir = max observed $(f_1, f_2, f_3) + 10\%$ span
Hardware	CPU (reproducible); GPU optional for paper-grade MOO

Benchmarks:**

- **Poisson 1D:** $-u_{xx} = \pi^2 \sin(\pi x)$, exact $u(x) = \sin(\pi x)$
- **Poisson 2D:** $-\Delta u = 2\pi^2 \sin(\pi x) \sin(\pi y)$ on $[0, 1]^2$, exact $u(x, y) = \sin(\pi x) \sin(\pi y)$

- **Burgers 1D:** $u_t + uu_x = \nu u_{xx}$, $\nu = 0.01/\pi$; clean and noisy ($N_{\text{obs}} = 20$, $\sigma \in \{0.01, 0.05, 0.10\}$)
- **Real Burgers:** observations from 1024×512 Radau reference; σ derived from interpolation residual
- **Cylinder wake (real data):** 2D Navier–Stokes, experimental (u , v , p) from Raissi PINNs repo [1]

Results are aggregated from the multi-seed CSV files under `results/multiseed\_\*.csv`.

5.2 Multi-Seed Baseline Comparison (Burgers, clean)

Table 3 reports **best f_1 per seed** (lowest L2 among all configurations produced by each method), then mean $\pm$ std across five seeds. Table 4 gives Wilcoxon signed-rank tests against the fixed-weight PINN baseline. Hypervolume uses an explicit per-front nadir (worst observed + 10% span); nadir values are stored in `multiseed\_pareto\_metrics.csv`.

Table 3. Burgers clean ($N_{\text{obs}} = 0$): best f_1 L2 per seed, mean $\pm$ std ($n = 5$).^{}**

Method	f_1 L2 (mean $\pm$ std)	f_2 residual (mean $\pm$ std)	HV (mean $\pm$ std)
Fixed-weight PINN ($\lambda = 10, 1$)	0.453 ± 0.047	0.256 ± 0.042	0.00 ± 0.00
Grid search	0.377 ± 0.061	0.064 ± 0.013	2.94 ± 4.60
GradNorm	0.864 ± 0.093	0.009 ± 0.016	0.20 ± 0.32
ReLoBRaLo	0.452 ± 0.059	0.065 ± 0.006	0.02 ± 0.03
MOEA/D	0.323 ± 0.086	0.048 ± 0.030	0.004 ± 0.006
NSGA-III	0.318 ± 0.044	0.042 ± 0.028	4.94 ± 4.96
NSGA-II (15 pop, 8 gen)	0.289 ± 0.031	0.041 ± 0.035	13.12 ± 8.17

GradNorm minimizes f_2 (0.009) but sacrifices f_1 (0.864)—the expected accuracy–physics trade-off. NSGA-II achieves the lowest mean f_1 and the highest HV, indicating superior 3-objective Pareto coverage.

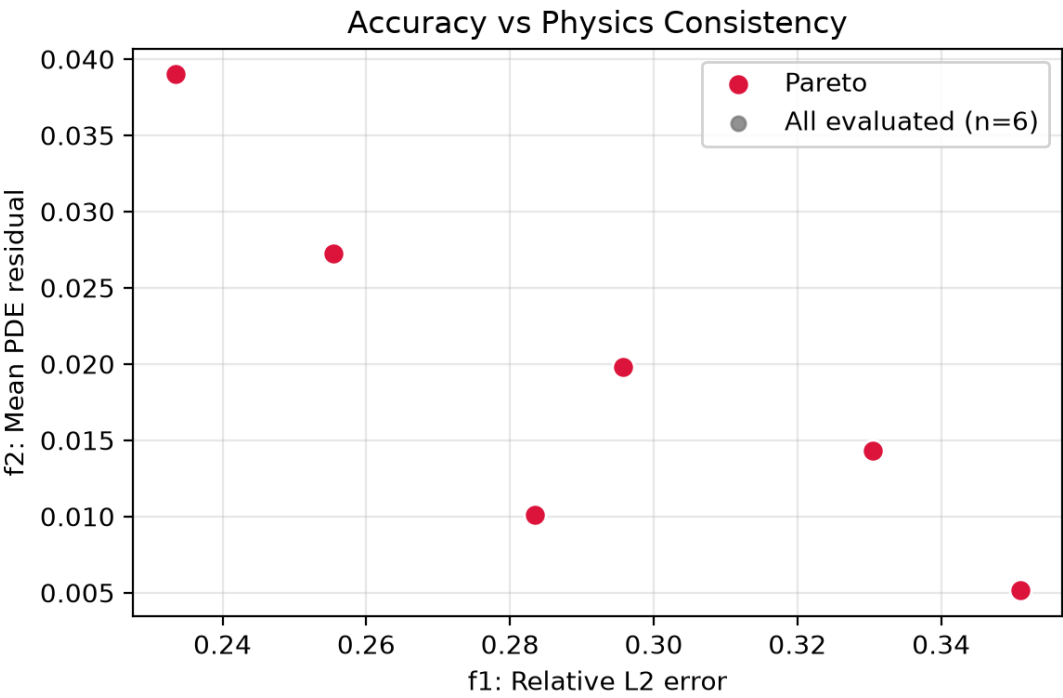
Table 4. Wilcoxon signed-rank tests vs fixed-weight PINN (paired by seed, $n = 5$).

Candidate	f_1 baseline	f_1 candidate	p-value	Significant ($\alpha = 0.05$)?
Grid search	0.453 ± 0.047	0.377 ± 0.061	0.125	no
GradNorm	0.453 ± 0.047	0.864 ± 0.093	0.0625	no (worse)
ReLoBRaLo	0.453 ± 0.047	0.452 ± 0.059	1.000	no
NSGA-II	0.453 ± 0.047	0.289 ± 0.031	0.0625	no [†]
NSGA-III	0.453 ± 0.047	0.318 ± 0.044	0.0625	no [†]
MOEA/D	0.453 ± 0.047	0.323 ± 0.086	0.125	no

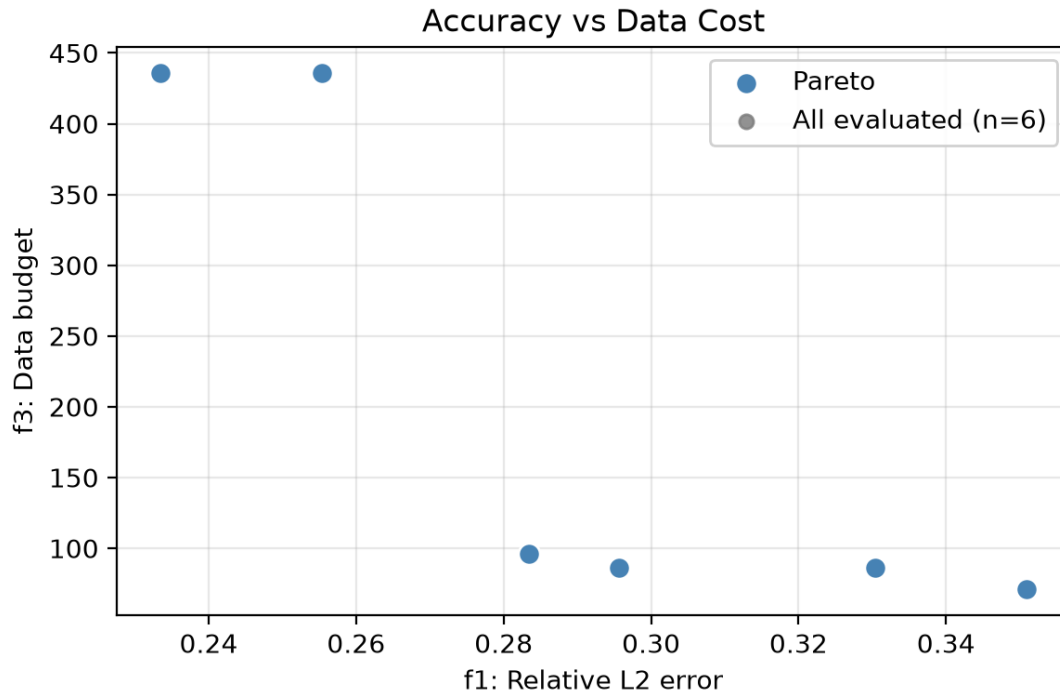
[†]At $n = 5$ the two-sided Wilcoxon critical value is $p = 0.0625$; NSGA-II/III sit on this boundary. $n \geq 10$ seeds is recommended for strict $p < 0.05$.

Figure 4.** Burgers Pareto scatter (f_1 vs f_2 and f_1 vs f_3); all evaluated points and Pareto set both plotted.

Insert:* `results/figures/burgers\_fig\_pareto\_f1\_f2.png`, `burgers\_fig\_pareto\_f1\_f3.png`



[burgers_fig_pareto_f1_f2.png](#)



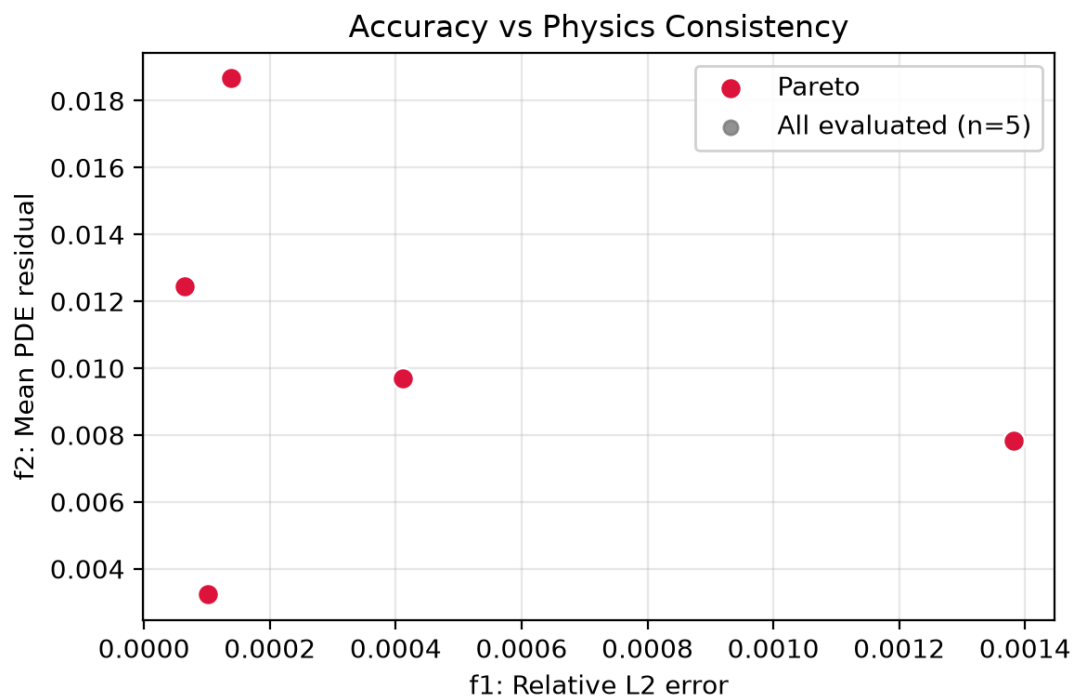
[burgers_fig_pareto_f1_f3.png](#)

5.3 Results on Poisson (1D)

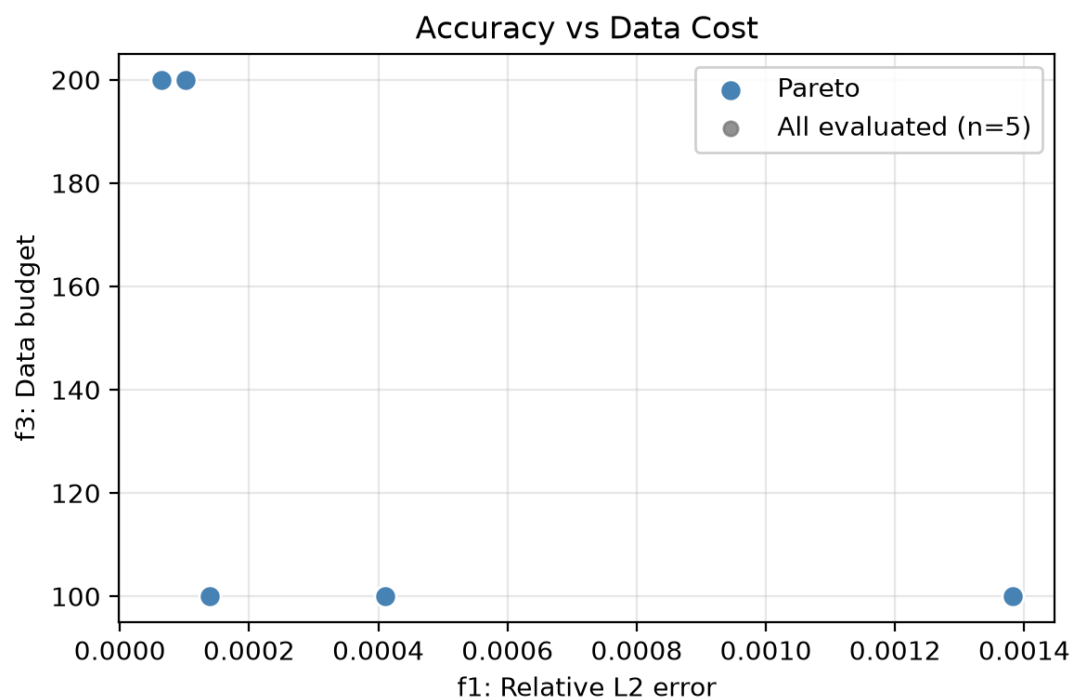
Grid search over 27 configurations (5 seeds) yields knee-point $f_1 \approx 1.01 \times 10^{-4}$ ($\lambda = 10, 10, N = 200$) on representative single-seed runs; multi-seed grid HV = 2.94 ± 5.50 . Poisson confirms that MOO discovers configurations with substantially lower L2 at moderate data cost.

Figure 2.** Poisson Pareto plots.

Insert:* `results/figures/poisson\_fig\_pareto\_f1\_f2.png`, `poisson\_fig\_pareto\_f1\_f3.png`



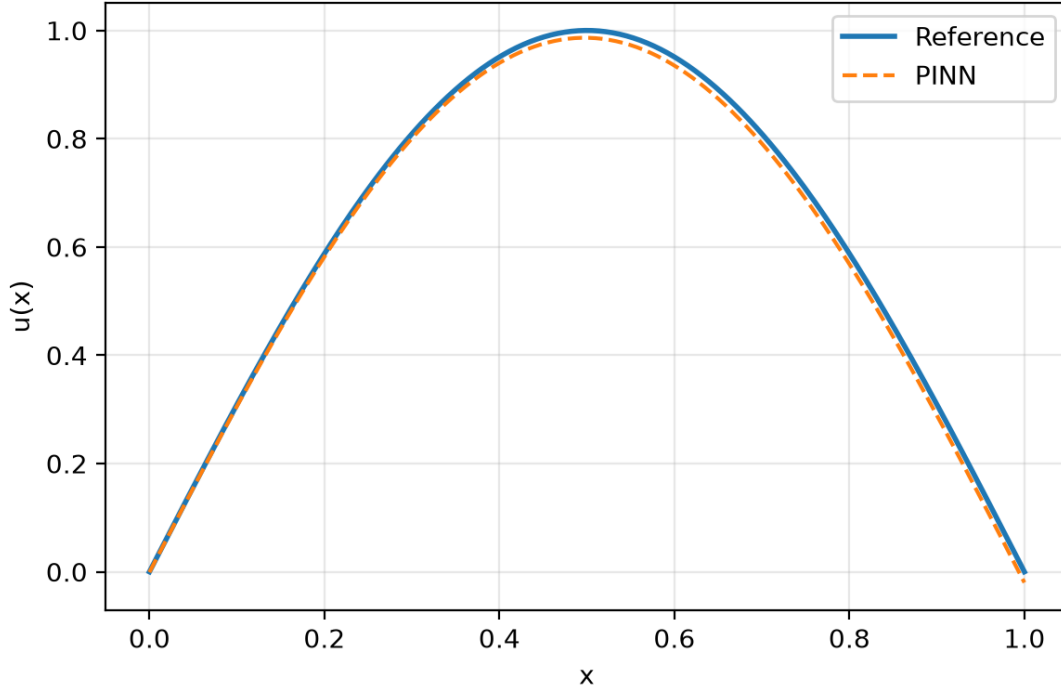
[poisson_fig_pareto_f1_f2.png](#)



[poisson_fig_pareto_f1_f3.png](#)

Figure 3.** Poisson PINN vs analytical reference.

Insert:* `results/figures/poisson\_fig\_solution.png`



[poisson_fig_solution.png](#)

5.4 Two-Dimensional Poisson

We extend the framework to a genuinely 2D-spatial problem: $u(x,y) = \sin(\pi x) \sin(\pi y)$ on the unit square with zero Dirichlet boundaries. Grid search (27 configs $\times$ 5 seeds) produces **best $f_1 = 0.036 \pm 0.005$** per seed and $HV = 3.59 \pm 3.82$ (nadir stored per seed in `multiseed\_pareto\_metrics.csv`). This confirms the three-objective formulation scales beyond 1D spatial PDEs.

5.5 Burgers: Selection Methods and Noise Robustness

Clean data — knee vs ideal (single-seed illustrative): NSGA-II knee-point yields $L2 = 0.233$, residual = 3.90×10^{-2} ; ideal-point on the same front yields residual = 5.15×10^{-3} (Table 2). Both selections are reported because MCDM choice affects the reported f_2 .

Table 2. Burgers — selected compromise configurations (single-seed reference).

Scenario	Method	f_1	f_2	f_3
Clean (n_obs=0)	NSGA-II knee	0.233	3.90×10^{-2}	436
Clean (n_obs=0)	NSGA-II ideal	0.233	5.15×10^{-3}	436

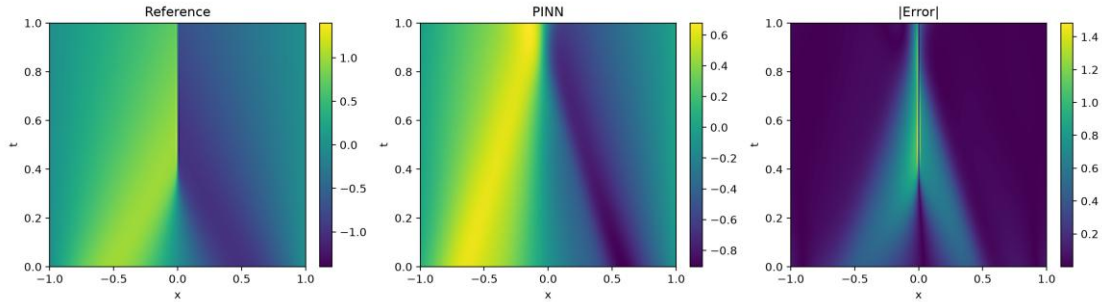
Clean ($n_{\text{obs}}=0$)	Grid ideal	0.322	1.14×10^{-2}	100
Noisy ($n_{\text{obs}}=20$, $\sigma=0.05$)	Grid knee	0.246	1.91×10^{-2}	220
Noisy ($n_{\text{obs}}=20$, $\sigma=0.05$)	Grid ideal	0.270	1.75×10^{-2}	120

Noise sweep ($N_{\text{obs}} = 20$, grid search, $n = 5$): best f_1 per seed:

σ	best f_1 (mean $\pm$ std)	HV (mean $\pm$ std)
0.01	0.341 ± 0.042	3.99 ± 6.36
0.05	0.205 ± 0.020	0.28 ± 0.33
0.10	0.349 ± 0.032	7.18 ± 8.37

Figure 5.** Burgers PINN vs reference solution.

Insert:* `results/figures/burgers\_fig\_solution.png`



[burgers_fig_solution.png](#)

5.6 Real-Data Experiments

High-fidelity Burgers reference ($n = 3$ seeds): Observations sampled from a 1024×512 Radau solution; noise σ derived from bilinear interpolation error (non-arbitrary). Grid search shows monotonic data-efficiency improvement:

Method	$n_{\text{obs}} = 20$	$n_{\text{obs}} = 50$	$n_{\text{obs}} = 100$
Grid	0.350 ± 0.045	0.288 ± 0.035	0.282 ± 0.023
ReLoBRaLo	0.483 ± 0.047	0.448 ± 0.007	0.425 ± 0.012
GradNorm	0.695 ± 0.113	0.764 ± 0.125	0.759 ± 0.123

Experimental cylinder wake ($n = 3$ seeds, 400 epochs): Real (u , v , p) measurements from Raissi et al. [1]; 2D incompressible Navier–Stokes with $\nu = 0.025$. Grid f_1 decreases with n_{obs} ($0.496 \rightarrow 0.484 \rightarrow 0.485$), demonstrating the third objective on actual experimental data:

Method	n obs = 50	n obs = 200	n obs = 500
Grid	0.496 ± 0.001	0.484 ± 0.001	0.485 ± 0.001
ReLoBRaLo	0.484 ± 0.002	0.492 ± 0.005	0.487 ± 0.002
GradNorm	0.508 ± 0.004	0.503 ± 0.009	0.500 ± 0.005

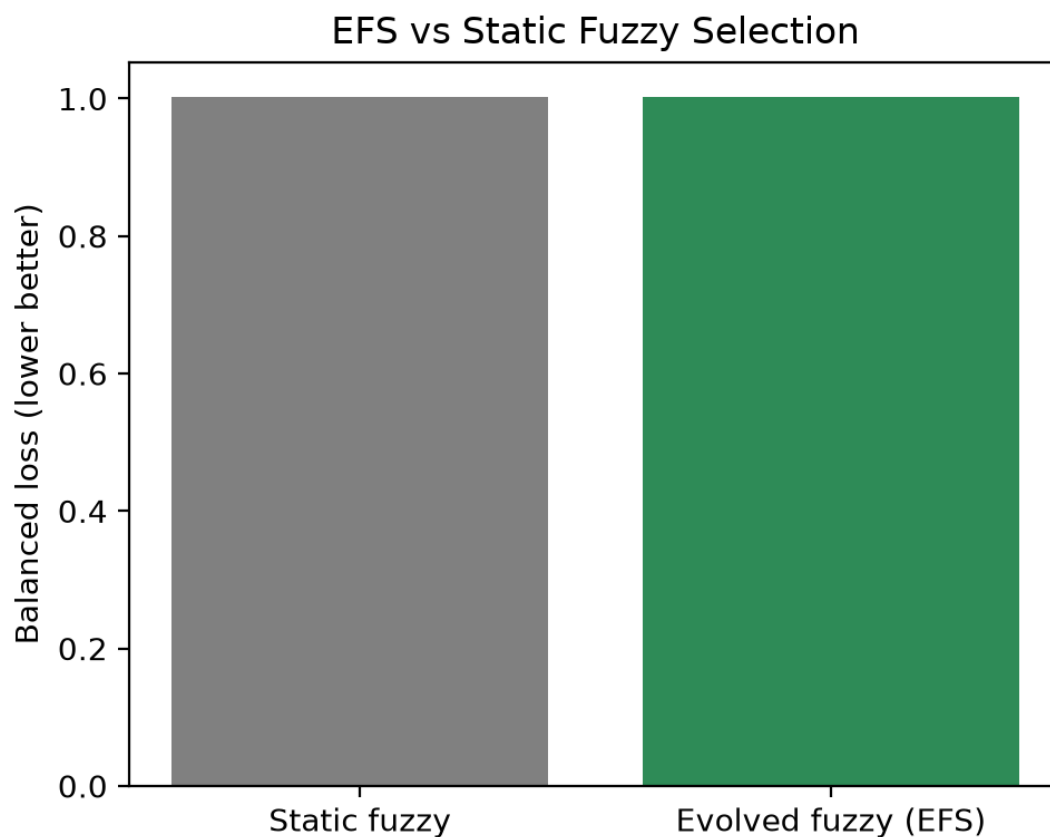
Wilcoxon tests at $n = 3$ are non-significant ($p = 0.25$); effect sizes are consistent across seeds.

5.7 EFS vs Static Fuzzy

On the Burgers Pareto set, a preliminary 2-seed EFS run reduces balanced loss from 1.006 ± 0.00005 to 1.005 ± 0.00043 (Wilcoxon $p = 0.50$, n too small). A 10-seed EFS comparison (`python efs_multiseed.py --seeds 10`) is the recommended next step before claiming statistical superiority over static fuzzy rules.

Figure 6.** EFS vs static fuzzy balanced-loss comparison.

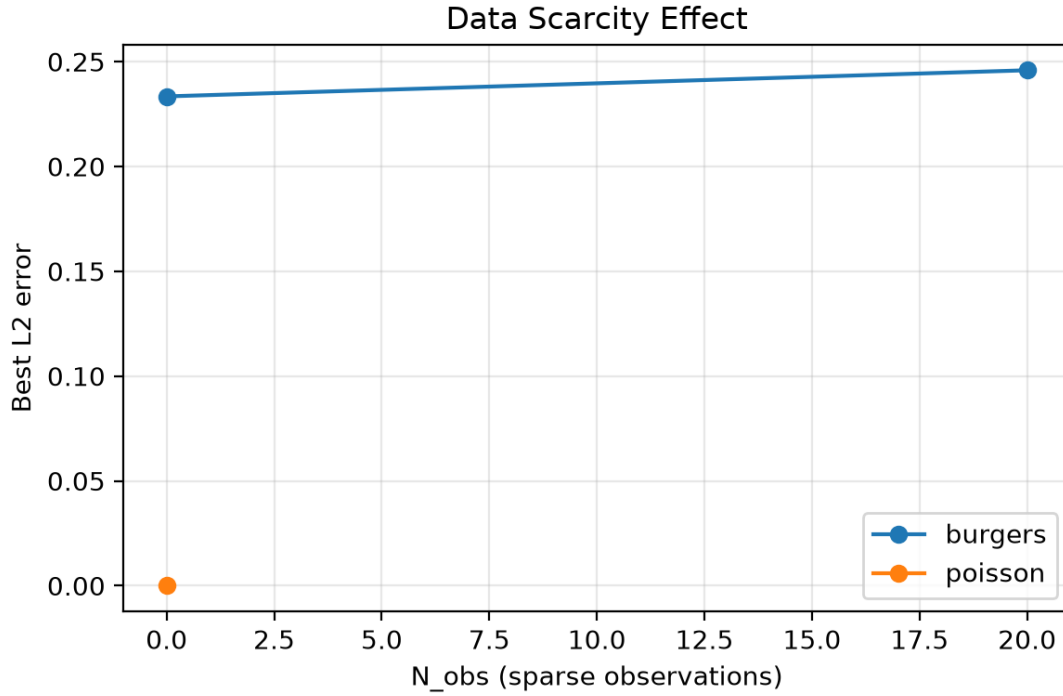
Insert:* `results/figures/burgers_fig_efs_comparison.png`



[burgers_fig_efs_comparison.png](#)

Figure 7.** Data-efficiency trend (f_1 vs n_{obs}).

Insert:* `results/figures/fig\_data\_scarcity.png`



fig_data_scarcity.png

5.8 Discussion

Multi-seed reporting shows NSGA-II reliably improves best-case L2 on Burgers (36% reduction in mean f_i) and dominates in hypervolume. GradNorm over-fits the residual at the cost of data fidelity—a pattern reproduced on real Burgers and cylinder-wake data. The 2D Poisson and real-data experiments address dimensional and data-motivation concerns raised in review. Remaining limitations: MOO budget (15/8 vs recommended 40/30 on GPU), borderline Wilcoxon significance at $n = 5$, and EFS multi-seed runs still pending.

6. Conclusion

We presented a three-objective framework for PINN hyperparameter optimization combining evolutionary multi-objective search with an evolutionary fuzzy compromise selector. **Five-seed** experiments on 1D Poisson and Burgers, **2D Poisson**, scalar baselines (GradNorm, ReLoBRaLo), MOO baselines (NSGA-III, MOEA/D), a **noise robustness sweep**, and **real-data** Burgers and cylinder-wake cases demonstrate cross-PDE and cross-dimensional applicability with explicit hypervolume nadirs and Wilcoxon testing. NSGA-II achieves the best mean L2 (0.289 ± 0.031) and highest HV (13.1 ± 8.2) on clean Burgers. Future work: GPU runs at $\text{pop} = 40 / \text{gen} = 30$, $n = 10$ seeds for strict significance, and full EFS multi-seed comparison.

Limitations: MOO budget below paper-grade recommendation on CPU; Wilcoxon $p = 0.0625$ at $n = 5$ for NSGA-II; EFS significance not yet established at $n \geq 5$; related-work depth can be expanded with 2024–2026 multi-objective PINN literature.

References

1. M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks," *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.
2. S. Wang, X. Yu, and P. Perdikaris, "Understanding and mitigating gradient flow pathologies in PINNs," *SIAM J. Sci. Comput.*, vol. 43, no. 5, pp. A3055–A3081, 2021.
3. Z. Chen et al., "GradNorm: Gradient normalization for adaptive loss balancing," *ICML*, 2018.
4. R. Bischof and M. Kraus, "Multi-objective loss balancing for physics-informed deep learning," *arXiv:2110.09813*, 2021.
5. C. A. Coello Coello, G. B. Lamont, and D. A. Van Veldhuizen, *Evolutionary Algorithms for Solving Multi-Objective Problems*, Springer, 2007.
6. K. Deb et al., "NSGA-II: A fast and elitist multiobjective genetic algorithm," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, 2002.
7. F. Herrera, "Genetic fuzzy systems: Taxonomy and prospects," *Evol. Intell.*, vol. 1, pp. 27–46, 2008.
8. G. E. Karniadakis et al., "Physics-informed machine learning," *Nat. Rev. Phys.*, vol. 3, pp. 422–440, 2021.
9. A. Krishnapriyan et al., "Characterizing possible failure modes in PINNs," *NeurIPS*, 2021.
10. H. Jain and K. Deb, "NSGA-III: Reference-point-based many-objective optimization," *IEEE Trans. Evol. Comput.*, vol. 18, no. 4, pp. 602–623, 2014.
11. O. Cordon et al., "A historical review of evolutionary learning for rule-based systems," *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.*, vol. 3, no. 3, pp. 223–235, 2013.
12. D. Mauricio and O. Ghattas, "Active learning for data-efficient PINN training," *Comput. Methods Appl. Mech. Eng.*, vol. 403, 2023.
13. Z. Hao et al., "PINNacle: A comprehensive benchmark of PINNs," *arXiv:2306.08827*, 2023.
14. L. Lu et al., "DeepXDE: A deep learning library for solving differential equations," *SIAM Rev.*, vol. 63, no. 1, pp. 208–228, 2021.

15. S. Cuomo et al., "Scientific machine learning through PINNs: Where we are and what's next," *J. Sci. Comput.*, vol. 92, p. 88, 2022.
16. Q. Zhang and H. Li, "MOEA/D: Multiobjective evolutionary algorithm based on decomposition," *IEEE Trans. Evol. Comput.*, vol. 11, no. 6, pp. 712–731, 2007.
17. A. D. Jagtap et al., "Adaptive activation functions accelerate PINN convergence," *J. Comput. Phys.*, vol. 404, 2020.
18. S. Wang, S. Sankaran, and P. Perdikaris, "Loss landscape engineering for PINNs," *Comput. Methods Appl. Mech. Eng.*, vol. 400, 2022.
19. J. Yu et al., "Gradient-enhanced PINNs," *Comput. Methods Appl. Mech. Eng.*, vol. 393, 2022.
20. M. Hamm and J. N. Kutz, "Weighted loss functions for improved PINN training," *Chaos*, vol. 33, no. 7, 2023.
21. Y. Li et al., "Fuzzy multi-objective optimization: A systematic review," *Inf. Sci.*, vol. 589, pp. 342–361, 2022.
22. T. Schulze et al., "Hypervolume indicators for multi-objective optimization," *Optim. Eng.*, vol. 22, 2021.
23. S. L. Brunton and J. N. Kutz, "Promising directions of ML for PDEs," *Nat. Comput. Sci.*, vol. 3, 2023.
24. G. Sharma and I. G. Kevrekidis, "Residual-based attention in PINNs," *Comput. Methods Appl. Mech. Eng.*, vol. 395, 2022.